BALWA MOHAMMADALI

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SUMMARY

Computer Engineering student with hands-on experience building machine learning and deep learning projects in Python, including image classification, NLP, and retrieval-augmented generation (RAG). Completed a Machine Learning internship, led a Smart India Hackathon 2025 team, and served trained models through FastAPI. Seeking entry-level AI/ML, Data Science, or Python roles.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering 2023 – 2027
Gujarat Technological University, Ahmedabad CGPA: 8.96 / 10

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL (MySQL)
Libraries & Frameworks: NumPy, Pandas, Matplotlib, Sk-Learn, Tensorflow/keras, Django, FastAPI, Pydantic
ML/DL: Supervised and Unsupervised Learning, NLP, ANN, CNN, RNN, LSTM, Transfer Learning
Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, Excel, n8n, Antigravity

PROJECTS

Plant Disease Detection System | EfficientNetB0, TensorFlow/Keras, FastAPI, Grad-CAM

- Built a 39-class plant disease image classifier using EfficientNetB0 transfer learning, with image preprocessing and augmentation.
- Fine-tuned the final 20 layers to reach 99.15% validation accuracy and ∼99% macro precision, recall, and F1-score across 11,105 validation samples.
- Added Grad-CAM visual explanations and served the trained model through a FastAPI prediction API for image-based inference.

RAG-Based AI Agent for PYQ Analysis | RAG, Gemini, Pinecone, n8n

- Built a RAG-based AI agent that retrieves and analyzes previous year exam questions by subject, chapter, marks, and exam trends.
- Automated document ingestion from Google Drive with n8n, split text using Recursive Character Text Splitting, and generated embeddings with Google Gemini.
- Stored embeddings in Pinecone and used context-grounded retrieval to identify frequently asked questions and generate exam-focused answers.

Movie Recommendation System | NLP, Scikit-Learn, FastAPI

- Built a content-based recommender by combining genres, keywords, cast, crew, and overview into a single feature set.
- Vectorized text features with CountVectorizer and generated recommendations using cosine similarity.
- Integrated the TMDB API to fetch movie posters and display results in an interactive web interface.

EXPERIENCE

Machine Learning Intern – Edunet Foundation | Remote May 2025 – Jun 2025

- Built a heart disease prediction model in Python, covering data preprocessing, feature engineering, model training, and evaluation.
- Performed EDA and visualization using Pandas/NumPy and trained classification models with Scikit-Learn.

HACKATHONS & COMPETITIONS

- **Smart India Hackathon 2025 (Team Lead):** Led a team to build a CNN-based cattle and buffalo breed recognition system using MobileNet, TensorFlow/Keras, and Django, achieving ∼74% validation accuracy with real-time prediction and Top-3 results.
- **UNESCO Global Youth Hackathon 2025:** Built an NLP and machine learning-based fake news detection system for real-time text classification and prediction.

CERTIFICATIONS

Python – HackerRank | Machine Learning – IBM SkillsBuild | Java Programming – Great Learning